PRODUCT STRATEGY | CASE STUDY

RideConvert

Urban Mobility Membership Growth

An end-to-end product strategy for converting casual bike-share riders into annual members. Validated through data analysis, user research, and strategic prioritization.

github.com/aashi2912/rideconvert rideconvert.vercel.app

SWIPE >>

THE PROBLEM SPACE

The Universal Bike-Share Revenue Challenge

70 - 80%

of rides come from casual pay-per-ride users

3 - 5×

more lifetime revenue from annual members

2 - 4%

industry-wide casual → member conversion rate

THE CORE INSIGHT

Casual riders don't convert because the product was not designed to make membership feel obviously worth it for their usage pattern. The value proposition assumes a commuter. Most casual riders are not commuters.

STRATEGY

Three Horizons. One Deliberate Constraint.

HORIZON	GOAL	SUCCESS METRIC
6 Months	Prove the nudge works	A/B test ≥25% lift
12 Months	Flexible membership tier	New tier ≥4% attach rate
24 Months	Membership majority	Members > casual volume

WHAT I EXPLICITLY DECIDED NOT TO BUILD

- **Price discounting**

  Trains users to wait for deals. Permanently destroys LTV. Will not ship.

- **Referral program**

  Acquisition tool. Conversion problem not yet solved. Wrong sequence.

- **Gamification**

  High effort, low confidence for this user segment and price point.

The strategic constraint: *I explicitly chose NOT to target new user acquisition. Conversion of existing engaged users has higher expected value,*

USER RESEARCH

Four Personas. Data-Derived, Not Assumed.

K-means clustering on 150K rides across duration, timing, day-of-week, bike type, and seasonality.
Labels came after the clusters — not before.

Weekend Explorer

42% OF CASUAL RIDES

BEHAVIOR

Leisure rides, weekends, parks & waterfronts.

PAIN POINT

Single passes feel expensive for spontaneous rides.

Lever: Flexible tier (~C$65/yr)

Reluctant Commuter

31% OF CASUAL RIDES

BEHAVIOR

Rush-hour weekdays, near transit hubs.

PAIN POINT

ROI uncertainty for C$105/yr membership.

Lever: Break-even calculator

Seasonal Visitor

18% OF CASUAL RIDES

BEHAVIOR

Tourist clusters, summer only, landmarks.

PAIN POINT

Annual membership is the wrong product.

Lever: Short-term visitor tier

E-bike Adopter

9% OF CASUAL RIDES

BEHAVIOR

Chooses e-bikes, longer distance, faster trips.

PAIN POINT

Paying 2x the per-minute rate vs members.

Lever: "Member: C$X/2" nudge

DATA ANALYSIS

What 150,000 Simulated Rides Revealed

RIDE DURATION

Casual rides are 2× longer — every day

Members avg 11 min (utility/commute). Casual avg 29 min (leisure). Two distinct distributions, not one segment with variance.

WEEKLY PATTERN

Members commute. Casuals explore.

Members peak Tuesday–Thursday. Casuals dominate Saturday–Sunday. Consistent across Divvy, Citi Bike, and Toronto.

HOURLY PATTERN

The 8am/5pm commuter signal

Members: twin spikes at 8am and 5pm. Casuals: gradual build, peak at 2–4pm. Post-ride nudge fires at 5pm for peak motivation.

SEASONALITY

Toronto: 16× August vs January

Most extreme seasonality of any major operator. Summer is the conversion campaign window with the highest nudge ROI.

Source: Synthetic data calibrated to Toronto Parking Authority 2024 Business Review · School of Cities BST Analysis

PRODUCT STRATEGY

Opportunity Solution Tree

GOAL: INCREASE CASUAL → ANNUAL MEMBER CONVERSION RATE

Remove price barrier	Personalize value prop	Reduce conversion friction
Flexible membership tier (~C$65/yr for Weekend Explorer)	Post-ride savings nudge (C$X spent → C$8.75/mo membership)	One-tap upgrade flow (6 steps → 2 steps)
Break-even ride calculator (24 rides = C$105 payback)	Usage-triggered emails (frequent casuals only)	Geo-targeted offer at top casual stations

A/B TESTING | USER INTERVIEWS | FUNNEL ANALYTICS | COHORT ANALYSIS

PRIORITIZATION

What We Build First - and Why

RANK	INITIATIVE	REACH	IMPACT	CONF.	EFFORT	RICE SCORE
#1	**Post-ride savings nudge**	HIGH	HIGH	HIGH	LOW	92
#2	**Flexible membership tier**	HIGH	HIGH	MED	MED	78
#3	**Break-even calculator**	MED	HIGH	HIGH	LOW	74
#4	**Geo-targeted station offer**	HIGH	MED	MED	LOW	61
#5	**One-tap upgrade flow**	MED	HIGH	MED	MED	54

WHY THE NUDGE IS #1

It intercepts riders at peak motivation - immediately after a ride (peak-end rule). Minimum engineering, maximum behavioral impact. No new infrastructure required. Works in-app or via SMS across all operators.

ROADMAP

3-Quarter Execution Plan

Quarter 1

QUICK WINS

- Post-ride savings nudge (in-app / SMS)
- Break-even calculator widget
- Funnel instrumentation & baseline

SUCCESS CRITERIA

≥20% lift in A/B test

Quarter 2

PERSONALIZE

- Usage-triggered emails (frequent casuals)
- Geo-targeted offers at top 15 stations
- CTA copy A/B testing (5 variants)

SUCCESS CRITERIA

Conversion rate 4.0%+

Quarter 3

NEW TIER

- Flexible membership SKU (~C$65/yr)
- One-tap upgrade flow (6 → 2 steps)
- Corporate employer partnerships

SUCCESS CRITERIA

Overall conversion 5.0%

METRICS

How We Know It's Working

NORTH STAR METRIC

Casual → Member conversion rate within 90 days

90 days captures seasonal riders without including noise. Measures product effectiveness independent of operator scale.

METRIC	CURRENT	Q3 TARGET
Conversion Rate	2.8%	**5.0%**
Members Added / Mo	3,500	**6,250**
Incremental ARR	Baseline	**+C$363K**
Time-to-convert	34 days	**21 days**

ACQUISITION

% casuals exposed
CTA click-through rate

ACTIVATION

Sign-up flow started
Drop-off per step

CONVERSION

Casual → member rate
Rate by persona

RETENTION

Year-2 renewal rate
Ride freq post-convert

GUARDRAIL METRICS: App uninstall rate · Casual rider NPS · Casual ride volume · Member year-1 churn

THE DELIVERABLES

End-to-End. Every Layer.

Not a slide deck. A working product built to production standards.

Live Analytics Dashboard

- React + Recharts deployed on Vercel
- 5 interactive views: Overview, Segments, Funnel, A/B, Roadmap
- Interactive A/B test simulator with statistical power feedback

Python EDA Notebook

- 21 cells, 7 sections using pandas & scikit-learn
- K-means clustering validates 4-persona model
- A/B test power analysis & sample size calculations

SQL Analysis

- 3 files, 1,091 lines of production-ready SQL
- Funnel analysis, nightly persona classifier, retention cohorts
- Compatible with BigQuery, Snowflake, and PostgreSQL

Strategy & Product Docs

- 6-page strategy brief with financial model
- Full PRD with user stories and launch criteria
- Decisions Log: 7 logged decisions with rationale

PRODUCT STRATEGY | USER RESEARCH | RICE PRIORITIZATION | ROADMAPPING | METRICS & KPIS | A/B TEST DESIGN | PYTHON | SQL | REACT